

- 2 For many years, farmers have used artificial insemination (AI) as a method to improve the milk yield of dairy cattle.

In cattle, sperm cells carrying the X chromosome contain 3.8% more DNA than the sperm cells carrying the Y chromosome. Female cattle have the sex chromosomes XX.

Assisted reproduction programmes allow dairy farmers to increase the number of female offspring produced.

One technique used to identify which sex chromosome a sperm cell contains is centrifugation.

Some semen samples were collected from male cattle.

- 10 cm³ samples of semen were placed in centrifuge tubes with 15 cm³ of a buffer solution added to each tube.
- The tubes were placed in a centrifuge and left to spin for 20 minutes at 1275 revolutions per minute.
- The spinning causes the sperm cells containing the X chromosome to fall to the bottom of the tubes. The sperm cells containing the Y chromosome remain in the fluid at the top.

Semen samples which have been processed in this way are referred to as sexed samples.

- (a) (i) State **three** variables which were standardised in this technique.

- 1
.....
.....
- 2
.....
.....
- 3
.....
..... [3]

- (ii) Suggest **one** reason why it is important that the semen sample is **not** spun at a very high speed.

-
.....
.....
..... [1]

Another method of sexing semen samples is to use a technique called flow cytometry as shown in Fig. 2.1.

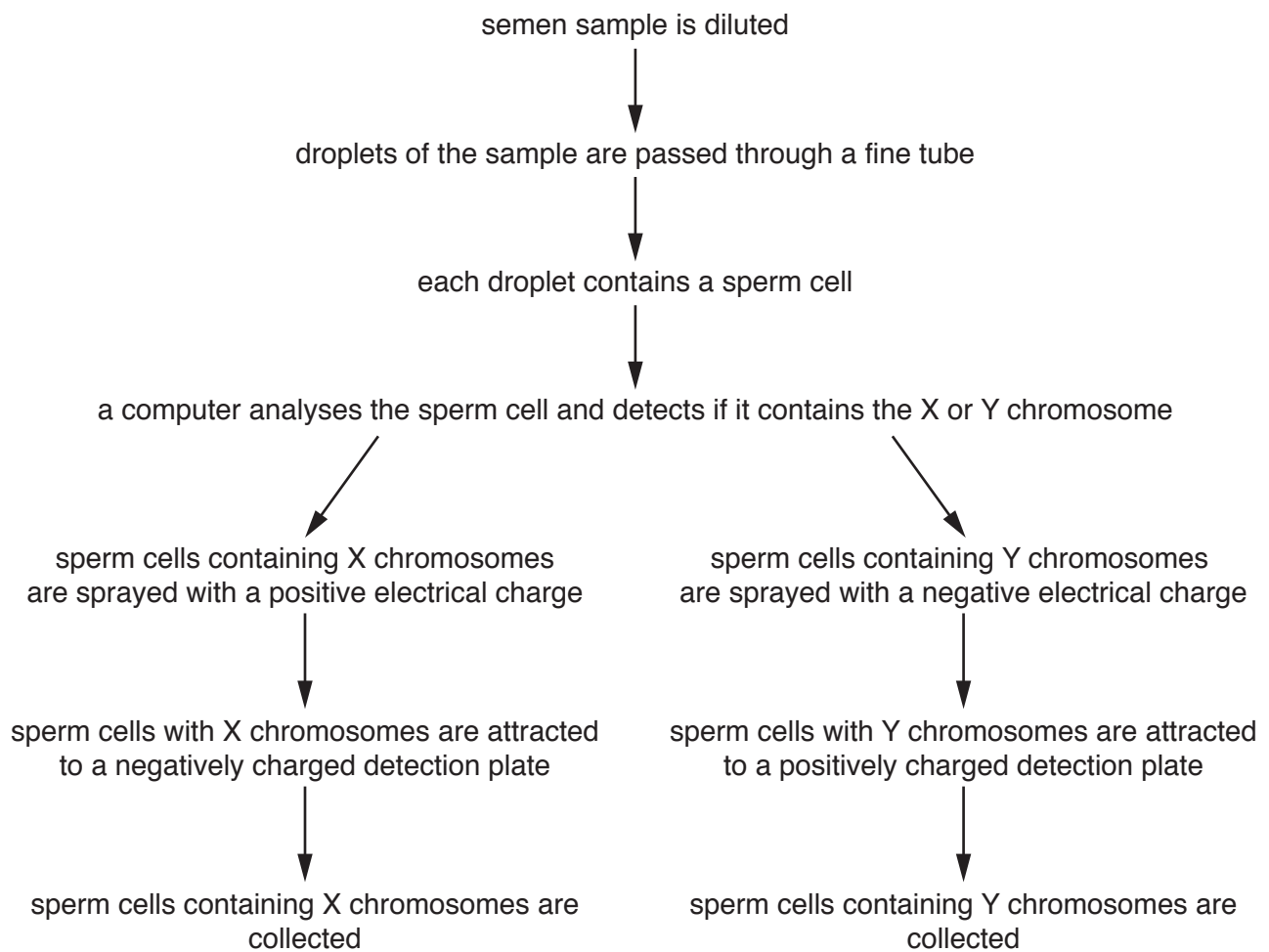


Fig. 2.1

The samples containing X chromosomes are frozen in small plastic straws, allowing semen samples to be stored for many years and transported all over the world.

- Flow cytometry can process 8000 sperm cells every second.
- One straw can hold 2.0×10^6 sperm cells.

(b) Calculate how many straws can be filled every hour by flow cytometry.

Number of straws filled every hour = [1]

The semen is analysed by technicians to determine its quality before it is sexed.

Technicians use microscopy to look at the appearance and motility of the sperm cells and to estimate the sperm count.

Fig. 2.2 shows the surface view of a haemocytometer used to estimate the number of sperm cells in a sample.

The depth of the haemocytometer is 0.1 mm.

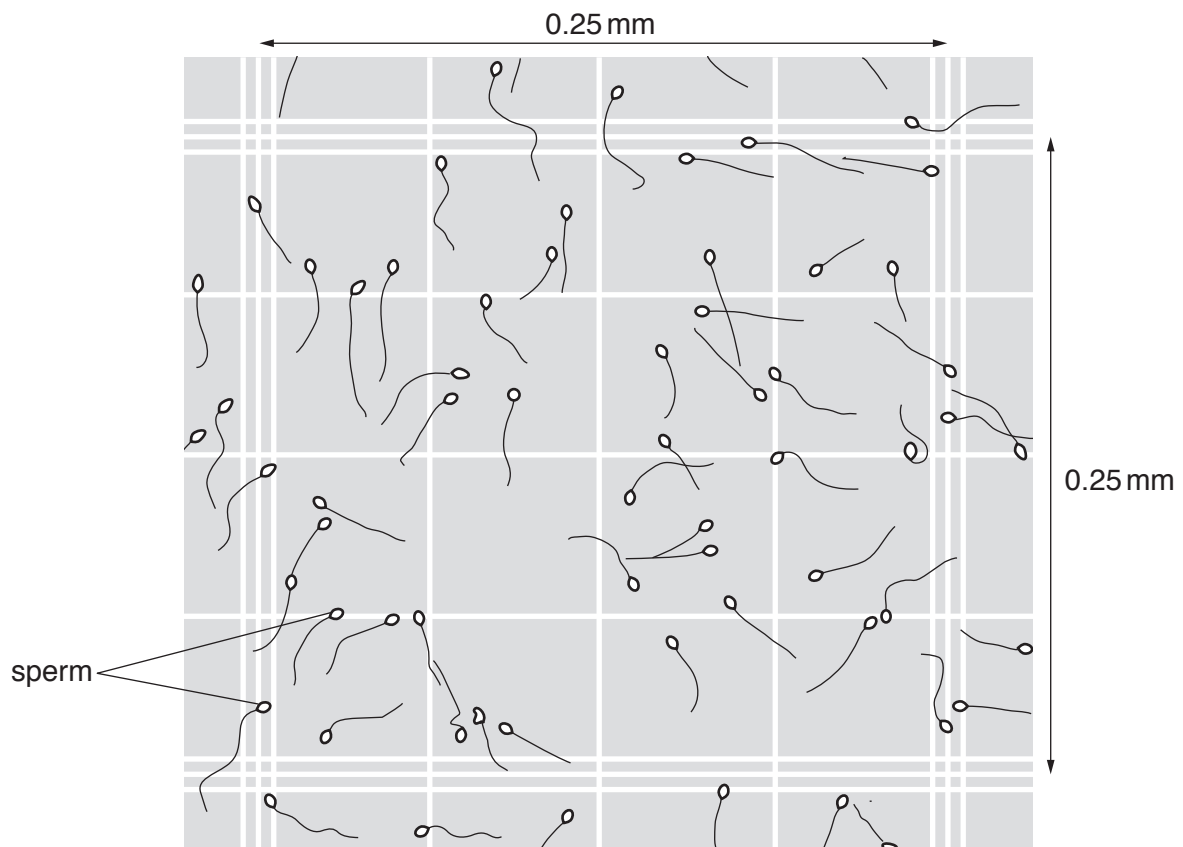


Fig. 2.2

- (c) Suggest how this apparatus could be used to estimate the number of sperm cells per cm^3 of semen **and** describe how the technician decided which sperm cells to include in the count.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [5]

[Total: 10]